Automatic Street Light /Dark Detector Circuit

ECE3570 Project

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What the Project Is About

This project is an automatic street light circuit that uses an LDR sensor and an op-amp to detect light intensity. The circuit turns the LED on when it becomes dark and turns it off when there is enough light.

Main idea:

-Detect surrounding light level automatically
-Compare the sensor voltage with a reference voltage
-Control the LED output based on light conditions

Purpose and Real-World Application

Purpose

- To design a circuit that automatically responds to changes in light conditions
- To reduce manual switching of lights
- To demonstrate how sensors and op-amps can be used in control systems

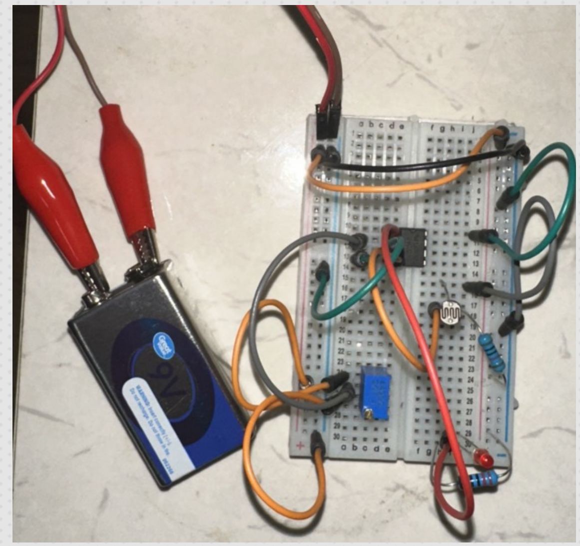
Real-World Application

- automatic street lighting
- garden and outdoor lights
- security lighting systems
- parking lot lighting
- energy-saving smart lighting systems

Components and Circuit Description

Components Used

- LM741 Op-Amp
- LDR (Light Dependent Resistor)
- 10kΩ Resistor
- Potentiometer
- 220Ω Resistor
- LED
- 9V Battery
- Breadboard and jumper wires



Circuit Description

- The LDR and 10kΩ resistor form a voltage divider
- The potentiometer sets the reference voltage
- The op-amp compares the LDR voltage with the reference voltage
- The LED turns ON or OFF depending on the light level

Benefits, Drawbacks, and Real-Life Implementation

Benefits

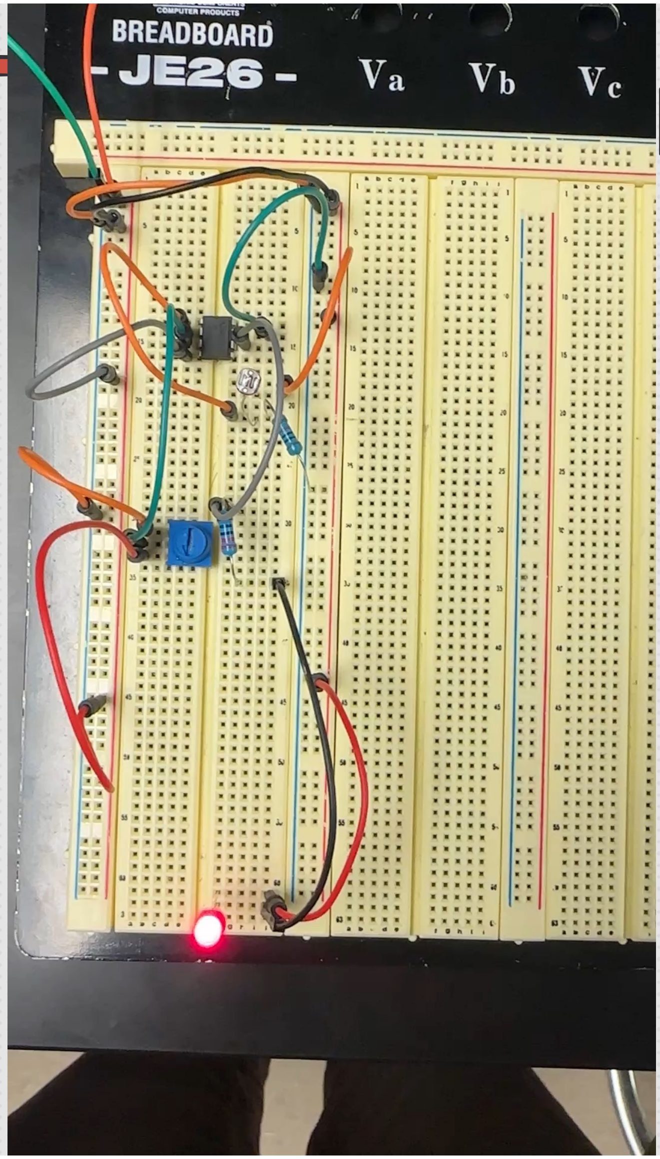
- simple and low-cost design
- automatic operation
- saves energy
- easy to build and understand
- useful for basic smart-lighting applications

Drawbacks

- sensitivity needs adjustment using the potentiometer
- performance can change with surrounding light conditions
- the LM741 is not the best op-amp for single-supply operation
- the circuit is suitable for small loads unless extra components are added

Real-Life Implementation

- can be connected to a transistor or relay to control larger lamps
- can be used in homes, pathways, campuses, and public lighting
- helps reduce power waste and improves convenience



Project Demonstration Video

This LDR sensor detects light intensity in the project and helps the circuit turn the LED on in darkness and off in bright light.

This video shows:

- how the circuit responds to light and darkness
- LED turning on in dark conditions
- LED turning off in bright conditions
- the effect of adjusting the potentiometer
- the working model in real time